

MECHANICAL DETERMINANTS OF GRADIENT WALKING ENERGETICS IN MAN

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SUMMARY

1. The metabolic cost and the mechanical work at different speeds during uphill, level and downhill walking have been measured in four subjects.

2. The mechanical work has been partitioned into the internal work (W_{int}), due to the speed changes of body segment with respect to the body centre of mass (BCM), and the external work (W_{ext}), related to the position and speed changes of the BCM in the environment.

3. W_{ext} has been further divided into a positive part (W_{ext}^+) and a negative one (W_{ext}^-), associated with the energy increases and decreases, respectively, over the stride period.

4. For all constant speeds the most economical gradient has been found to be -10.2% (± 0.8 S.D.).

5. At each gradient there is a unique $W_{\text{ext}}^+/W_{\text{ext}}^-$ ratio ($= 1$ in level walking), regardless of speed, with a tendency for W_{ext}^- and W_{ext}^+ to vanish above $+15\%$ and below -15% gradient, respectively.

6. W_{int} is constant at each speed regardless of gradient. This is partly explained by an only slight decrease in stride frequency at increasing gradient. W_{int} constancy implies that it has no role in determining the optimum gradient.

7. A linear multiple regression relating W_{ext}^+ and W_{ext}^- to the metabolic cost at different gradients showed that negative (eff^-) and positive (eff^+) efficiencies decrease with increasing speed (from 0.912 to 0.726 , and from 0.182 to 0.146 , respectively). The $\text{eff}^-/\text{eff}^+$ ratio, however, remains rather constant (4.995 ± 0.125 S.D.).

8. We conclude that the measured W_{ext} , the $W_{\text{ext}}^+/W_{\text{ext}}^-$ partitioning and $\text{eff}^-/\text{eff}^+$ ratio, i.e. the different efficiency of the muscles used as force and brake generators, can explain the metabolic optimum gradient at about -10% .

INTRODUCTION

In a classic paper Margaria (1938), by measuring the metabolic demand of locomotion at different gradients, concluded that at an incline of about -10% the energy cost of walking is minimum. Due to the lack of adequate technology for motion analysis it was not feasible to investigate the mechanical determinants of this energetically optimal gradient for walking. The mechanical work was estimated to be

proportional to the incline angle, hence Margaria considered only the minimum changes of potential energy needed by the progression. In spite of this approximation, Margaria obtained, from metabolic and mechanical results of walking below -20% and above $+20\%$ (the range associated with a linear behaviour of the cost of

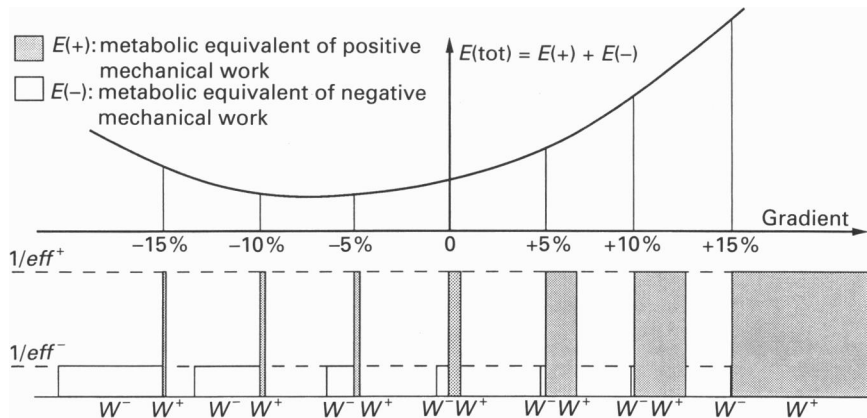


Fig. 1. The positive and negative component (W^+ and W^- , widths of the bottom rectangles) constitutes one-half of the total mechanical work at level walking (0 gradient). We assume that to maintain the same speed uphill and downhill at different gradients both the total work increases and the contribution of one of its components becomes prominent (positive-uphill, negative-downhill). By multiplying the rectangle bases by the proper heights (which corresponds to dividing by the efficiency of positive and negative work), we obtain areas proportional to the metabolic equivalent of each component of the mechanical work (stippled and white patterns). The cumulative area for each gradient, given as the ordinate in the upper graph, has an asymmetric profile with a minimum corresponding to a certain gradient in the downhill zone.

locomotion), reliable first evaluations of the efficiency of human locomotion (and indirectly of muscular contraction).

It is well known (Williams & Cavanagh, 1983) from physics that in level walking at constant speed the amount of positive mechanical work, mostly due to ascending and accelerations of the body centre of mass (BCM), must be counterbalanced by an equal quantity of negative work, related to descendings and to a decelerative, 'braking' effect. This is caused by the periodic nature of locomotion, both in the vertical ('level' gait) and horizontal ('constant' average speed) direction. When walking downhill or uphill, the partitioning will no longer be 50% of the total, and we expect the negative and positive work to prevail, respectively, in the two conditions. This behaviour matches the 'rimless (polygonal) wheel' similarity suggested for the human locomotion machinery (McGeer, 1990). In fact, moving such a wheel on horizontal ground requires equal quantities of acceleration and deceleration. In addition, the non-uniform progression still needs small amounts of braking action (particularly at heel-down) during uphill walking and a small amount of acceleration (just before toe-off) in downhill gait. In contrast, rolling a 'normal' wheel uphill or downhill at constant speed will require only positive or negative work, respectively.

While some studies have appeared on the effects of gradient in human locomotion (Davies & Barnes, 1972; Davies, Sargeant & Smith, 1974), there have been no reports of the positive-negative partition of mechanical work in gradient walking. It is nevertheless conceivable that a suitable combination of positive-negative mechanical work, together with the different efficiency values reported for positive and negative muscular work (Abbot, Bigland & Ritchie, 1952), could be responsible for the metabolic minimum at about -10% gradient. Figure 1 shows how this hypothesis could work (see legend).

The purpose of this study was to measure metabolic cost and mechanical work of gradient walking and to explain the determinants of the optimum gradient of about -10% .

METHODS

Four healthy male volunteers (aged 26.3 ± 7.5 years, mass 73.0 ± 7.2 kg, stature 1.81 ± 0.10 m) walked on a motor-driven treadmill with different gradients (from -15 to $+15\%$, step 5%) at speeds of 1.08 , 1.33 , 1.61 and 1.86 m s $^{-1}$. In each condition simultaneous mechanical and metabolic measurements were taken.

Biomechanical measurements

Each speed/gradient trial was analysed by means of an opto-electronic motion analysis system (ELITE System, B.T.S., Italy) with four cameras sampling at 100 Hz. Then reflective markers were placed on each side of the body in order to locate the subject's segments of interest: the head-trunk (ear lobe-iliac crest), thigh (great trochanter-knee joint), shank (knee joint-ankle malleolus), foot (heel-toe), upper arm (shoulder-elbow) and forearm (elbow-wrist). The 3D position of segment extremities was stereometrically calculated by the apparatus. Each acquisition lasted 5 s and the stride frequency was calculated by analysing the trace periodicity of the vertical co-ordinates *vs.* frame. The correspondence between the real treadmill speed and the imposed one was assessed by the analysis of foot markers during the contact time. The slippage was assumed to be negligible since the rubber belt was driven by a rubber cylinder with high friction between the two surfaces.

Each segment mass, centre of mass position and radius of gyration was taken from standard tables (Dempster, Gabel & Felts, 1959). Positive internal work (W_{int}) was computed by the method of Cavagna & Kaneko (1977). In order to account for the kinetic energy changes of segments whose movements do not affect the position of the common centre of mass, typically the symmetric limb displacement, they used the König theorem from mechanics. The theorem states that the total kinetic energy of a multi-link system can be partitioned into two parts: (a) the kinetic energy of segments coming from the change of their speed with respect to the overall centre of mass, and (b) the kinetic energy of the overall centre of mass. The second term is included in external work (W_{ext}) computation, while the first one has been named internal work, W_{int} . Our implementation of their method resulted in the following differences: (1) the present values refer to a 3D, 2-sided co-ordinate system, (2) the head-trunk segment is included as a separate moving part of the König theorem, (3) the reference point for calculation of segment relative speeds is here the true body centre of mass, obtained by the position of all twelve body segments, (4) the translational and rotational kinetic energies of adjacent limb segments were added together, allowing for a within-limb energy transfer.

The 3D displacement of the body centre of mass was calculated for each frame from the position of the twelve body segments. Positive W_{ext}^+ and negative W_{ext}^- were obtained by summing the increments and the decrements, respectively, of the total energy (= potential + horizontal kinetic + vertical kinetic) curve of the body centre of mass *vs.* time. Total external work (W_{ext}) was computed as the sum of the previous two terms.

All the data processing and statistics have been performed on an Olivetti M380 XP5 and an Apple Macintosh Quadra 700 computer. Low-pass filtering of the data was performed using the procedure of D'Amico & Ferrigno (1990), in which optimal filter cut-off frequencies are automatically selected. The range of cut-off frequencies was 5.0–9.0 Hz using this procedure. A

custom program, created using LabView 2/Macintosh (National Instruments, Austin, TX, USA) developing environment, analysed the 3D data and calculated the biomechanical parameters (112 strides were measured). The procedure for data differentiation, needed for marker speed computing, was the straight dx/dt , dy/dt and dz/dt calculation on the filtered co-ordinates.

Metabolic measurements

A standard open-circuit technique was used. O_2 and CO_2 partial pressures were sensed by gas analysers (OM-11 and LB-2, Beckman Inc., Schiller Park, IL, USA) and their values, together with a signal proportional to the displacement of a spirometer bell, were fed into a MINC 11-23 computer (Digital Inc., Marlboro, MA, USA) for STPD conversion of ventilation and O_2 uptake calculation. Laboratory temperature was always within 21–24 °C. Standing data were subtracted from the steady-state values, reached after about 4 min of walking, to obtain the net energy expenditure at each gradient/speed trial.

Mechanical and metabolic work values in the present paper are expressed per kilogram of body weight and per metre travelled.

RESULTS

The optimum gradient, at which the metabolic cost is minimum, was estimated from coefficients of second-degree polynomial regression for each subject and speed, as reported in Table 1. It was found to be -10.2% and the small standard deviation, 0.8% , suggests that it is not affected by the speed.

TABLE 1. Regression coefficients of equation $E = a + b \times \text{gradient} + c \times \text{gradient}^2$ for each walking speed, where E is expressed in $\text{ml } O_2 \text{ kg}^{-1} \text{ km}^{-1}$ and gradient as a percentage. The optimum gradient, according to calculus, has been computed as $-b/(2c)$

Speed (m s^{-1})	<i>a</i>	<i>b</i>	<i>c</i>	<i>r</i> ²	Optimum gradient (%)
1.08	114.850	9.385	0.435	0.969	-10.79
1.33	110.870	10.114	0.516	0.989	-9.80
1.61	122.570	10.329	0.560	0.994	-9.22
1.86	150.780	10.771	0.490	0.985	-10.99

Conversely, W_{int} data, shown *vs.* gradient in Fig. 2, were found to be rather constant with respect to gradient and mainly dependent on speed. Figure 3 reports the fractions of the external mechanical work due to positive and negative work as a function of gradient. The small s.d. values indicate that the partitioning is not affected by speed.

DISCUSSION

With respect to the search for the mechanical determinants of the metabolic cost of gradient walking, particularly in relation to the minimum occurring at a gradient of about -10% , the role of W_{int} appears not to be relevant. Its constancy, at each speed, cannot affect the W_{ext} shape since it constitutes an offset in the work *vs.* gradient graph. Furthermore, by definition, its value (positive internal) must be counterbalanced by an equal amount of negative internal work (50% of the total internal work, regardless of gradient), provided that the limb speed change is considered with respect to the position of the body centre of mass (BCM). Such an average limb speed has to equal the overall progression speed, as has been shown recently by a theoretical model of bipedal walking (Minetti & Saibene, 1992), which

shows that W_{int} is a linear function of speed and stride frequency (as obtained by expressing their equation A21 per unit distance, like the measurements reported in Fig. 2). In our experiments we noticed that at each speed the stride frequency tends to decline slightly with increasing gradient only during uphill walking. In this

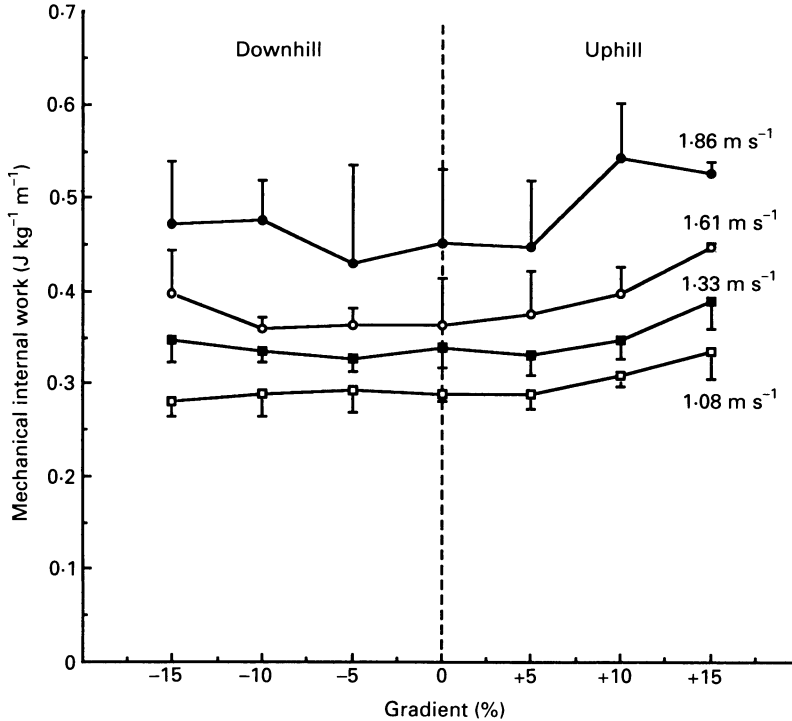


Fig. 2. Internal work per unit distance at different gradients (average of all subjects). Standard errors ($n = 4$) have been used for graphical purposes.

perspective, the discrepancy between the lower frequency and a small W_{int} increase can be regarded as being produced by the assumed kinetic energy transfer between adjacent limb segments.

By following the hypothesis suggested in Fig. 1 about the mechanical determinants of the optimum gradient of -10% , both the relative values of W_{ext}^+ and W_{ext}^- and the efficiency of positive (eff^+) and negative (eff^-) work play a crucial role. When the indicated procedure of converting the mechanical work into its metabolic equivalent is used to simulate the effect of such variables on the optimum gradient, it emerges that important factors are: (a) the $W_{\text{ext}}^-:W_{\text{ext}}^+$ ratio (as a percentage of W_{ext}) which differs from $100:0$ for downhill gradients and $0:100$ for uphill gradients, and (b) the shape of this relationship *vs.* gradient. Figure 3 shows the experimentally obtained sigmoidal pattern which we found, by means of simulation with an electronic spreadsheet, to be better than a linear one (interpolated between the lower and upper limits) in determining the gradient of minimum metabolic cost (the symmetry of $W_{\text{ext}}^-:W_{\text{ext}}^+$ partitioning about the 0% gradient is an expected feature; see the legend of Fig. 4).

In addition it can be shown by simulation that changing either eff^+ or eff^- can affect the shape of the metabolic prediction as a function of gradient (at each constant speed), but if the ratio eff^-/eff^+ is kept constant, *ceteris paribus*, the predicted optimum gradient does not change. Since different efficiencies are expected

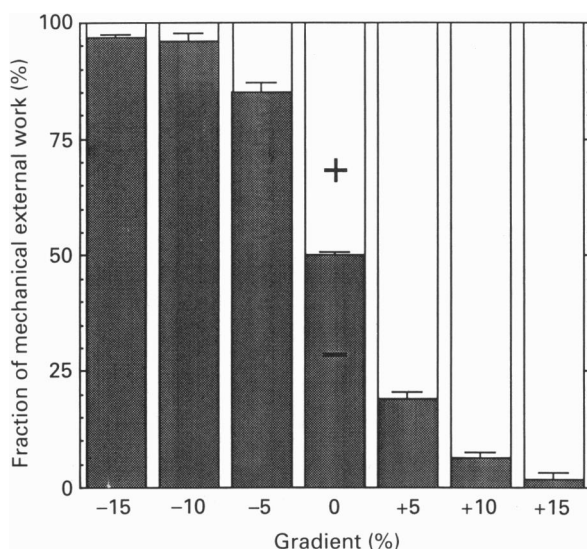


Fig. 3. Partitioning of the external mechanical work (100%) into positive (white) and negative (grey) subunits at different gradients. All speeds pooled. Vertical lines refer to S.D.

to occur at different contraction speeds (Kushmerick & Davies, 1969), and provided that we obtained experimentally the minimum metabolic cost at a constant gradient regardless of speed, an attempt to derive statistically eff^+ and eff^- was made to verify the simulation results. We estimated the two efficiencies by assuming that W_{ext} is the only component of the total mechanical work (allowed by the constancy of W_{int} , as stated above).

At each speed a multiple regression analysis was performed on the model equation:

$$C = \frac{W_{ext}^-}{eff^-} + \frac{W_{ext}^+}{eff^+},$$

where C is the measured metabolic cost of walking. This model implies that eff^+ and eff^- remain constant at all gradients within a certain speed. The analysis showed a decrease of both efficiencies at increasing speed (see Table 2) with an average eff^-/eff^+ ratio of 4.995 (± 0.125 S.D.). These values agree with the results obtained for pure positive and negative work reported by Aura & Komi (1986a, b). They also did not take into account W_{int} when evaluating the mechanical work done by using a 'sledge ergometer'. The efficiency values could both have been higher, approaching the values found by Margaria (1938) and by Margaria, Cerretelli, Aghemo & Sassi (1963) at extreme gradients, if W_{int} had been included as a cost predictor in the model

equation. The high eff^-/eff^+ ratio (about 5) reflects the results published by Abbot *et al.* 1952) from two subjects pedalling one against the other and supports our rationale about the optimum gradient.

We summarize in Fig. 4 how the obtained mechanical results can explain the gradient with minimum metabolic cost at an intermediate speed (the graphs for other

TABLE 2. Efficiency of positive and negative external work obtained with a multiple linear regression (at each speed $n = 28$)

Speed (m s ⁻¹)	eff^- (S.E.M.)	eff^+ (S.E.M.)	eff^-/eff^+
1.08	0.912 (0.104)	0.182 (0.005)	5.011
1.33	0.901 (0.088)	0.175 (0.004)	5.149
1.61	0.785 (0.087)	0.162 (0.005)	4.846
1.86	0.726 (0.043)	0.146 (0.005)	4.973

speeds, not shown here, lead to the same conclusions). The figure shows that at a constant speed there is a gradient range, namely from -10 to $+10\%$, where the measured W_{ext} , due to the residual contribution of W_{ext}^+ in downhill and of W_{ext}^- in uphill walking, is different from the minimum one needed for BCM displacement (represented at each gradient by the height of the stippled areas in Fig. 4; see legend for details). Such a deviation from the ideally expected situation, when converting the mechanical work into metabolic equivalents, is responsible for a cost (vertical distance from highest to lowest lines in the lower panel) with an average minimum at -8.44% ($\pm 0.47\%$ S.D.) gradient. This value has been calculated for all speeds from $eff^- = 1.25$ and $eff^+ = 0.25$ (ratio = 5). We used these values from Margaria's work but, as stated above, only the ratio between them, *ceteris paribus*, is responsible for a certain optimum gradient.

From the depicted elaboration it emerges also that the present data and those obtained by Margaria (1938) are complementary to each other. Margaria, working in a very wide range of gradients ($\pm 40\%$), found that below -20% and above $+20\%$, the walking efficiency was about 1.25 and 0.25, respectively. He did not measure the mechanical work of walking and an approximate estimate, here indicated as the minimum mechanical external work due only to the potential energy changes (W_{min} , see legend of Fig. 4), was used to estimate the different efficiencies. While this worked well outside the $\pm 15\%$ range of gradients, for the intermediate values the mechanical work estimates declined towards an unrealistic zero (corresponding to level walking). Our measurements confirm an exclusive contribution of W_{ext}^- below -15% and of W_{ext}^+ above -15% , but the combination of them found within that range shows that the efficiencies of 1.25 and 0.25 can apply to all gradients. The idea that locomotion efficiency was similar to that of the isolated muscle (for positive work) was suggested by Margaria (1938). More recently the prevalent view (Cavanagh & Kram, 1985) is that the 'overall' efficiency is the product of two components: (a)

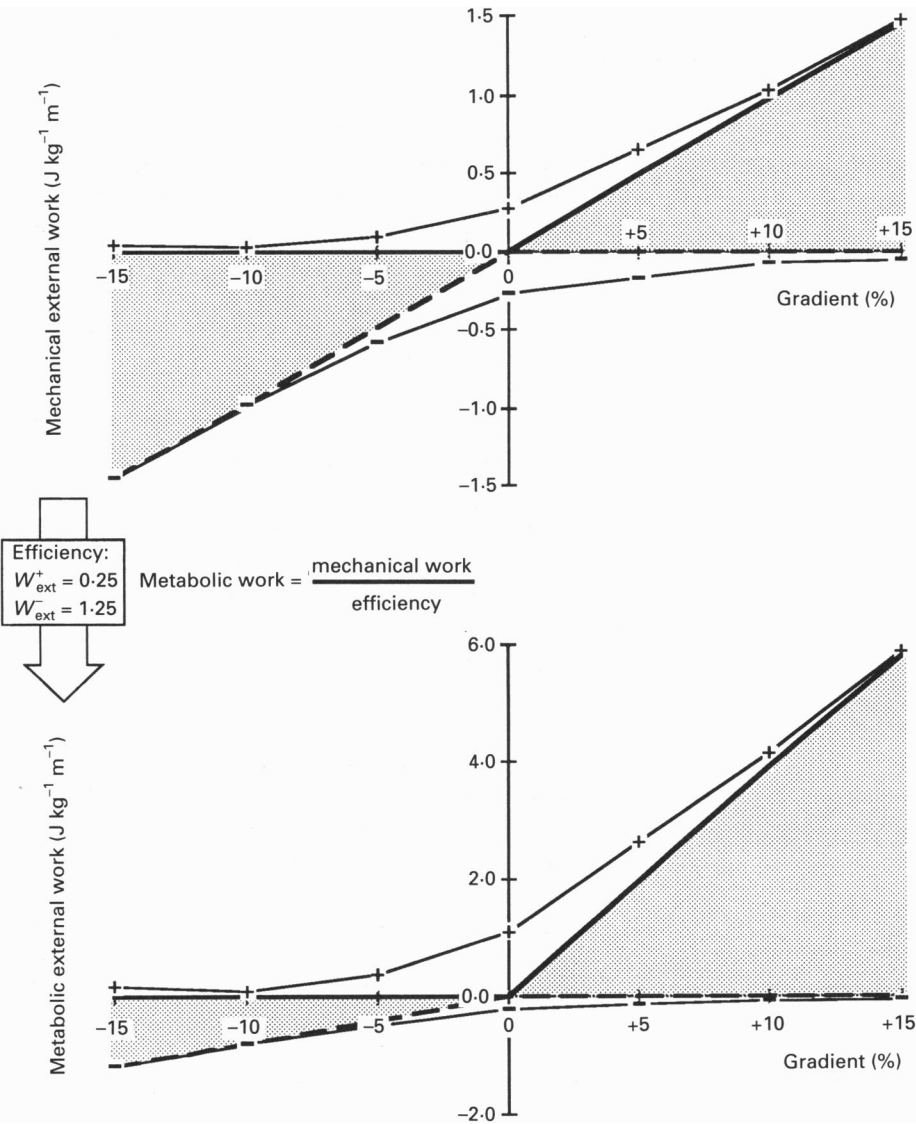


Fig. 4. Mechanical explanation of the minimum energy expenditure in downhill walking. Upper panel, mechanical external work as measured by means of motion analysis; lower panel, metabolic external work obtained by dividing the upper tracings by positive and negative efficiencies (the efficiency of the negative work has been considered with the positive sign for graphic purposes, see below). Both panels show positive (plus symbols) and negative (minus symbols) external work *vs.* gradient; thick lines correspond to the minimum external work (positive, continuous lines; negative, dashed lines) needed for gradient progression at constant (instantaneous) speed obtained according to the following expression:

$$\dot{W}_{\min} = \frac{\Delta \text{PE}}{\Delta t} = mg \frac{\Delta h}{\Delta t} = mg \bar{s}_{\text{vert}} = mg \bar{s} \sin(\tan^{-1} i),$$

muscle efficiency, as obtained in isolated preparations, and (b) the efficiency which accounts for the actuators arrangement in the musculo-skeletal system, for the interface between the human machine and the environment, and for the capacity of elastic structures to store and deliver energy (e.g. Ker, Bennet, Bibby, Kester & Alexander, 1987). Indeed, the validity of Margaria's suggestion is supported both by the highly conservative pattern of motion related to walking (high-energy 'recovery': Cavagna, Thys & Zamboni, 1976) and by the freely chosen walking pattern adopted by this experimental protocol which is expected to minimize the influence of co-activation of antagonist muscles and isometric contractions in determining a low 'transmission' efficiency (elsewhere referred to as 'effectiveness'; Lafortune & Cavanagh, 1983). In addition, in contrast to running, the elastic storage and release do not play a substantial part as a confounding factor (Williams & Cavanagh, 1983) in the mechanical work assessment of walking.

We are aware, though, that the assumption of constant efficiency at all gradients, at constant speed, can be regarded as a first approximation in the attempt to unveil the mechanical determinants of the optimum gradient. Another potentially misleading assumption is the partition of the total mechanical work into W_{int} and W_{ext} , as suggested by Cavagna & Kaneko (1977). While being designed to account for the movements of the limbs not affecting the BCM position (particularly those with BCM as the centre of symmetry), the separation between the kinetic energy of limbs from that of the BCM appears to cancel out all the energy saving due to ballistic oscillations of segments (Alexander, 1989), resulting in an overestimation of the total mechanical work. According to this expectation, we found that, assuming $\text{eff}^- = 1.25$ and $\text{eff}^+ = 0.25$, the metabolic equivalent work obtained including W_{int} overestimates the measured metabolic work in the majority of cases. However, we decided to use this method since: (a) the method of Cavagna & Kaneko (1977) has been demonstrated to be a good predictor of the work minimization, both theoretically

where $\dot{W}_{\text{min}}$ is the work rate related only to potential energy changes (ΔPE) during a stride period (Δt), Δh is the vertical displacement, m is the subject's mass, and $\dot{s}_{\text{vert}}$ and $\dot{s}$ are the vertical and the real progression speeds of BCM, respectively, along the incline with gradient i . Since for small angles

$$\sin(\tan^{-1} i) \approx i,$$

and the ordinate in the graphs is expressed as work per kilogram of body weight per metre travelled, the equation for the minimum external work turns into:

$$W_{\text{min}} = g\dot{i}.$$

Obviously, minimum positive and negative work have to be nil when moving downhill and uphill, respectively, at constant speed since only gravity has to be opposed (as could occur when riding an ideal bicycle). Looking at the upper panel, we can observe that at each gradient any positive excess with respect to W_{min} must be counterbalanced by an equal amount of negative work excess, in analogy with the 50 % partitioning between W_{ext}^+ and W_{ext}^- occurring in level walking. In the graphs, at each gradient value the total external work necessary to walk at a certain speed (1.33 m s^{-1}) is represented by the vertical distance between the two thicker lines (minimum work) and between the thinner ones (measured work). While in the upper panel the measured work is minimum at 0% gradient, such a symmetry (about 0%) is lost when the positive and negative work are converted into metabolic equivalent by using proper efficiency values (lower panel), revealing a vertical distance between the thin lines which is minimum near to -8% gradient.

(Minetti & Saibene, 1992) and experimentally (Cavagna & Franzetti, 1986), (b) the uncertainty about which other model to adopt is directly related to the variability in the mechanical work estimation and on the assumptions made (Williams & Cavanagh, 1983), and (c) it can be useful, as in this case, to work on an index, W_{ext} , reflecting the overall body performance. In addition, by looking at the energetics of both walking and running, the shape of the metabolic cost *vs.* speed is better approximated by the W_{ext} curve rather than the W_{int} one (Cavagna & Kaneko, 1977, their Fig. 3). It is likely that W_{int} is an important component at increased stride frequency, as in high-speed locomotion or when unnatural frequencies have been imposed on the subject (Cavagna & Franzetti, 1986).

In conclusion, we consider that the mechanics of uphill and downhill walking elucidate the relationship between the constraints associated with the human locomotor machinery and its metabolic needs. The minimum cost at about -10% gradient depends on the differences between the ideal energetic requirements (W_{min} lines in Fig. 4), due to exclusive potential energy changes, and the real ones. This work excess (the white zone between thick and thin lines in Fig. 4) originates from the mixing of acceleration (W_{ext}^+) and braking effect (W_{ext}^-) on the body centre of mass and their different efficiencies.

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